Construction Chemicals

Product Datasheet
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Jointing and Tiling

Back-fill

Closed-Cell Polyethylene
Foam Backer Rod for Joints



Product Description

Back-fill is a flexible, closed-cell polyethylene foam backer rod installed behind sealants in expansion and control joints. It controls the sealant depth, gives the correct sealant profile and acts as a bond breaker to prevent three-sided adhesion. It does not absorb water and is compatible with most joint sealants.

Uses

- Expansion and control joints in concrete and masonry.
- Joints in façades, curtain walls and precast panels.
- Floor joints before sealing.
- Joints around windows and doors.
- Backing for Mixseal PU and other O MIX sealants.

Advantages

- Controls sealant depth and profile.
- Natural bond breaker; prevents three-sided adhesion.
- Closed cells: no water absorption.
- Flexible; recovers more than 90 %.
- Remains flexible down to -40 °C.

Application

1. **Select Diameter**
   Choose a rod diameter so it is compressed to about 70 % of its diameter in the joint (40 % compression under water pressure).
2. **Insert**
   Press the rod into the joint with a blunt tool to a uniform depth; do not puncture or stretch it.
3. **Seal**
   Apply the sealant over the rod at the specified width : depth ratio.

Consumption

1 m per 1 m of joint
rod diameter selected to joint width
Do not stretch the rod during installation.

Packaging

Rolls in various diameters, or profiles up to 30 mm thick × 2 m long.

Storage & Shelf Life

Store in a dry, shaded place away from direct sunlight, heat sources and sharp objects.

Technical Data

Typical values at 25 °C

Property	Typical Value	Test Method
Material	Closed-cell polyethylene foam	—
Density	35 – 40 kg/m³	—
Colour	Blue / black	—
Tensile strength	70 kPa	—
Elongation at break	116 %	—
Compression recovery	> 90 %	—
Flexibility	down to -40 °C	—
Thermal conductivity	0.043 W/m·K	—
Water absorption	Negligible (closed cell)	—

Consumption

Joint width	Sealant depth
6 – 12 mm	Equal to joint width
12 – 24 mm	12 mm
> 24 mm	Half of joint width

Consumption is theoretical and varies with substrate porosity, profile and application method.

Application Conditions

- Joint faces clean, dry and sound before installing the rod.
- Rod compressed to about 70 % of its diameter (40 % under water pressure).
- Install at a uniform depth along the joint.
- Do not puncture, twist or stretch the rod.

Application Notes

- Use a blunt roller or wooden tool to insert the rod; never a sharp tool.
- Apply primer to joint faces only, not to the rod, where the sealant requires a primer.
- Apply the sealant soon after installing the rod.
- Butt-join rod ends; do not overlap.

Health & Safety

- Wear gloves when handling and cutting.
- Keep away from heat sources and open flames (combustible material).
- Follow site safety rules for work at height.

First Aid & Cleaning

- Eyes: rinse with clean water if particles enter the eye.
- Cuts: clean and dress the wound.
- Burns from contact with melted foam: cool with water and seek medical advice.
- No cleaning required; collect off-cuts in waste bins.

The information in this datasheet is based on laboratory tests and practical experience and is given in good faith. Users should verify product suitability for their specific application. O MIX reserves the right to update this information without prior notice.